Harness Engineering

Hackspree

Outline

- Introduction
- Principles of Harness Engineering
- Tooling and Automation
- Case study
- Summary

Introduction

Harness engineering is the practice of creating the scaffolding, tooling, and processes that let teams safely iterate on complex systems.

Principles

- Observability-first: make failures visible
- Incremental safety: small, reversible changes
- Automated validation: tests and canaries

Tooling and Automation

- CI pipelines tailored for harness tests
- Feature flag frameworks
- Replayable deployment orchestration

Case study

A short example of using harnesses to rollout a config change across services with canary monitors.

Summary

Harness engineering reduces blast radius and increases deploy confidence.

Appendix: Full post — Harness Engineering Best Practices

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title: Harness Engineering Best Practices
date: 2026-04-02
slug: harness-engineering-best-practices-for-ai-agents
summary: Strong agent systems depend on strong harnesses: repeatable tasks
tags: ai-agents, evaluation, harness
---

Agent quality is rarely limited by model intelligence alone. Most failures

If the harness is sloppy, the team ends up debating anecdotes instead of in

## Treat the harness as product infrastructure

A good agent harness is not a throwaway script. It is the system that tells

That means the harness should:
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Appendix: Full post — Coding Harnesses Need Real Repos

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---
title: Coding Harnesses Need Real Repos
date: 2026-04-11
slug: coding-agent-harnesses-need-real-repositories
summary: Coding-agent quality becomes measurable when the harness uses actu
tags: harness, coding-agents, swe-bench
---

[SWE-bench: Can Language Models Resolve Real-World GitHub Issues?](https:/

That is also the right pattern for harness design inside product teams.

A credible coding harness should prefer:

- messy repositories over perfect examples,
- failing tests over vague grading,
- issue-driven tasks over isolated snippets,
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Appendix: Full post — Harnesses Need Real Browsers

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---
title: Harnesses Need Real Browsers
date: 2026-04-19
slug: harnesses-need-real-browsers-not-polite-demos
summary: Web agent evaluation gets more useful when the harness runs agains
tags: harness, agents, browser
---

If an agent claims it can use the web, the harness should make it prove it

That is why [WebVoyager: Building an End-to-End Web Agent with Large Multi

The harness lesson is straightforward:

- use real interfaces,
- preserve the messy interaction sequence,
- score outcomes with concrete checks.
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Appendix: Full post — Harnesses Need Tasks That Fight Back

```
---
title: Harnesses Need Tasks That Fight Back
date: 2026-04-27
slug: agent-harnesses-need-tasks-that-fight-back
summary: Strong assistants are easier to trust when they are tested on tasl
tags: harness, gaia, evaluation
---

[GAIA: a benchmark for General AI Assistants](https://arxiv.org/abs/2311.1

That is a good model for internal harnesses too.

The harness should ask the system to do tasks that are:

- small enough to score,
- rich enough to require multiple steps,
- messy enough that shortcuts stop working.
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Appendix: Full post — Harnesses Should Observe the OS

```
---
title: Harnesses Should Observe the OS
date: 2026-05-03
slug: good-harnesses-watch-the-whole-operating-system
summary: Desktop and multimodal agents need execution harnesses that see tl
tags: harness, osworld, multimodal
---

The right environment can make an average agent look impressive or make a s

[OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Compu

That is the bar harness engineers should aim for. If your agent acts inside

- window state,
- clipboard and file effects,
- long action sequences,
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Appendix: Full post — Harness Engineering: Go for Reliable Pipelines

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---
title: Harness Engineering: Go for Reliable Pipelines
date: 2026-03-24
slug: go-is-good-for-harness-pipelines
summary: Harness engineering is fundamentally about building repeatable, t
tags: golang, harness, evaluation
---

Harness engineering is often less about model math and more about repeatabl

[SWE-bench: Can Language Models Resolve Real-World GitHub Issues?](https:/

That plumbing usually includes:

- work queues,
- sandbox orchestration,
- artifact capture,
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Appendix: Full post — Task Harness Engineering

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---
title: "Task Harness Engineering: how it compares to Eval and Agent Harnes
date: 2026-05-17
slug: task-harness-engineering
summary: "A practical guide to building task harnesses and how they differ
tags: [harness, evaluation, agents]
---

Task harnesses are the scaffolding that turn a high-level engineering ques
"Can this system finish a real task?"-into reproducible, debuggable experi

[![Harness Engineering talk (YouTube)](https://img.youtube.com/vi/C_GG5g38

Source: Harness Engineering (YouTube), referenced for conceptual framing a

What is a harness?
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References

- Harness Engineering Best Practices
- Coding Harnesses Need Real Repos
- Harnesses Need Real Browsers
- Harnesses Need Tasks That Fight Back
- Harnesses Should Observe the OS
- Harness Engineering: Go for Reliable Pipelines
- Task Harness Engineering